

Optimal trading strategies for multi-energy microgrid cluster considering demand response under different trading modes: a comparison study

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Abstract:

The linkage of neighborhood multi-energy microgrids (MEMGs) may overcome the disadvantage of one single microgrid (MG) since surplus/deficient energy can meet balance internally and realize self-satisfaction. In order to realize successful cooperation of such a complicated energy system, optimal trading strategy and reasonable trading price among all entities are important. In this study, a bi-level optimization model combined of two trading modes is proposed under different scenarios: intermediary agent (IA)-based trading mode where IA acts as an intermediate operator, as well as direct trading mode without IA. Stackelberg game theory (SGT) method and supply/demand (S/D) ratio method are used for price determination, respectively. Again, integrated demand response (IDR) is also considered in this study. As an illustrative example, a comparative study is implemented in a neighboring energy cluster combining three park-level MEMGs. The results show that, compared with the situation that each MEMG acts alone, a cluster obviously increase total benefit which may contribute to the development of tailored policy instruments. In addition, although direct trading among MEMGs enjoys higher benefit, self-sufficient ability is lower than the situation considering IA, which may help policy makers to more accurately predict merits and demerits associated with different energy trading modes among MEMGs.

Keywords: multi-energy microgrid, trading strategy, supply/demand ratio, stackelberg game theory, integrated demand response, intermediary agent.

Nomenclature

Abbreviations

<i>A</i>	Size of PV panel (m ²)	<i>Nch/Ndis</i>	Charging/discharging efficiency
<i>a</i>	Benefit coefficient	<i>nel/nht</i>	Internal electric/heat exchange of MEMG (kW)
<i>Bch/Bdis</i>	Battery charge/discharge power (kW)	<i>P</i>	Power (kW)
<i>Be</i>	Benefit (CNY)	<i>PCol</i>	Predicted cooling load (kW)
<i>BESS</i>	Battery energy storage	<i>Peb/Pes</i>	Internal electricity buy-back/sale price without IA (CNY /kWh)
<i>C</i>	Energy storage operating cost factor (CNY/kWh)	<i>PEle</i>	Predicted electric (kW)
<i>CHP</i>	Combined heat and power	<i>Pg</i>	Natural gas price (CNY /m ³)
<i>CNY</i>	Chinese Yuan	<i>PHet</i>	Predicted heat load (kW)
<i>col</i>	Optimized cooling load (kW)	<i>PrEb/PrEs</i>	Electricity price purchased from/sold to external network (CNY /kWh)
<i>COP</i>	Refrigeration coefficient	<i>PrHb/PrHs</i>	Heat price purchased from/sold to external heat network (CNY /kWh)

<i>Cpc</i>	Capacity of electric chillers (kW)	<i>Pueb/Pue</i> <i>s</i>	Utility electric buy-back/sale price (CNY /kWh)
<i>DERs</i>	Distributed energy resources	<i>PV</i>	Photovoltaic
<i>DES</i>	Distributed energy system	<i>Q</i>	Heat production of boiler (kW)
<i>DR</i>	Demand response	<i>r</i>	Heat to electric ratio
<i>elb/els</i>	Electric purchased/sold by MEMG (kW)	<i>S/D</i>	Supply/demand
<i>elde</i>	Electric demand of MEMG (kW)	<i>SGT</i>	Stackelberg game theory
<i>ele</i>	Optimized electric (kW)	<i>WT</i>	Wind power
<i>elsurp</i>	Total power supply of MEMG (kW)	<i>Binary variables</i>	
<i>Epeb/Epes</i>	Electricity price purchased/sold out by MEMG (CNY /kWh)	<i>keb,kech,kedis,kes,khb,k</i> <i>hs</i>	0/1 variables
<i>Es</i>	Stored power of the battery (kWh)	<i>Greek letters</i>	
<i>Etra</i>	Electric trade amount with utility grid (kW)		
<i>Exeb/Exes</i>	Electric purchased	<i>η</i>	Energy production

	from/sold to utility grid		efficiency
	(kW)		
<i>het</i>	Optimized heat load (kW)	<i>Subscripts</i>	
	Heat price purchased/sold		
<i>Hphb/Hphs</i>	out by MEMG (CNY	<i>bt</i>	battery
	/kWh)		
<i>htb/hts</i>	Heat purchased/sold by	<i>ec</i>	Electrical chillers
	MEMG (kW)		
<i>Htra</i>	Thermal energy transaction	<i>gb</i>	Gas boiler
	with external network (kW)		
<i>Hxhb/Hxhs</i>	Heat purchased from/sold	<i>h</i>	Hours
	to main network (kW)		
<i>IA</i>	Intermediary agent	<i>hc</i>	Heat exchanger
<i>Ii</i>	Irradiation (kW/m ²)	<i>max/min</i>	Maximum/Minimum
<i>k</i>	Supply/demand ratio	<i>mt</i>	Micro-gas turbine
<i>L</i>	Low calorific (kWh/m ³)	<i>n</i>	Number of MEMG
<i>M</i>	Gas consumption (m ³)	<i>ng</i>	Natural gas
<i>MEMG</i>	Multi-energy microgrid	<i>wh</i>	Waste heat boiler
<i>MILP</i>	Mixed integer linear	<i>wt</i>	Wind turbine
	Programming		

1. Introduction

Energy industry is undergoing a tremendous transformation from a centralized to a decentralized paradigm to reduce energy transmission losses and improve energy supply reliability. In order to deal with the intermittent problem of solar photovoltaic (PV)/wind power output within a distributed energy system (DES), microgrid (MG) [1-2] has been proved to be an ideal platform to integrate various distributed energy resources (DERs), which may enhance the utilization of renewable energy in a local distribution network. Along with the integration of various energy types such as electricity, heat, gas etc., MG has extended from pure electric linkage to multi-type energy (electricity/heat/gas) interchange and named as multi-energy microgrid (MEMG) [3-4]. Since few kinds of energy supply equipment and customer with similar load profiles lead to inflexible problem both for supply and demand sides in a single MEMG, it is ideal to combine nearby MEMGs to form a cluster, where the MEMGs may share energy with each other.

In this cluster, each MEMG needs to decide its local strategy on how to maintain supply/demand (S/D) balance itself, as well as market strategy on how to interact with neighborhoods and external public grid. Peer-to-Peer (P2P) energy trading represents such kind of direct trading mode, where energy is traded among local energy prosumers and consumers which is permitted according to current policy in China. To be specific, there are two kinds of trading modes: intermediate agent (IA)-based trading mode and direct trading mode among the MEMG operators. The former agent-based one may protect each member's privacy and simplify trading schedule, where IA may collect external public energy price, energy surplus/deficient amount information of each MEMG, and trade with each MEMG and external grid as an intermediate operator. Whereas, as an independent entity, IA aims at deriving price

difference profit between bid and ask. On the other hand, in the direct trading mode without IA, each MEMG trades with neighborhoods itself without IA's help. At present, China is carrying out a new round of energy mechanism reform, energy market is gradually opened up, and the energy pricing mechanism also needs breakthroughs. Under this background, energy trading forms will be more flexible and trading prices can be set more freely [5-6].

Furthermore, in order to realize interaction of energy supply-demand strategy and dynamic trading price, demand response (DR) cannot be ignored which may realize peak shaving and load shifting, thus enhance the flexibility of energy systems. Again, the demand side management may also affect the internal trading price since it may change the energy supply-demand situation in each time period. All in all, in such a complicated multi-level, multi-agent and multi-type energy network, rational planning and scientific decision-making are extremely challenging.

1.1. Literatures review

Many studies have tackled MGs with electric and other types of resources [7-8]. Moser et al. [9] proposed a modular energy management system for urban multi-energy systems. Guo et al. [10] presented a two-stage optimal planning and design method for combined cooling, heat and power (CCHP) MG system, multi-objective genetic algorithm based on non-dominated sorting genetic algorithm-II (NSGA-II) was applied. Gu et al. [11] reviewed the modeling, planning and optimal energy management of CCHP-based MG system. Moretti et al. [12] presented an innovative mixed integer linear programming (MILP) formulation for the operational planning of Multi-Energy Systems. Salata et al. [13] presented a procedure for the optimization of existing buildings using genetic algorithms. However, most of the studies focused on one single energy system, which can only achieve the local optimization of its internal

energy network, but cannot overcome the physical limitations such as equipment limitation and single load type.

To overcome this problem and realize energy sharing among the MGs, Bracco et al. [14] described a MILP model to optimally design and operate an energy system in a limited urban area where several buildings were connected by a heat distribution network. Although optimal energy supply and exchange strategy were discussed in this study, benefit allocation and energy price determination problem were not considered. Along with the appearance of novel energy ecosystem, energy trading problems within energy supply-demand systems must also be resolved. As early in 2009, Hawkes et al. [15] has indicated the importance of discussions on the benefit allocation among the stakeholders participated in energy systems, but only remained in the qualitative description level. Liu et al. [5] proposed a novel dynamic pricing methodology offering a market-oriented approach to drive decentralized energy trading problem and optimize financial benefits for DERs owners. Bui et al. proposed a multistep hierarchical optimization algorithm based on a multiagent system for energy management of multi-MGs systems [16]. In recent years, more and more studies utilize game theoretical strategies to deal with energy management problems and identified to be effective and feasible. Both cooperative and non-cooperative methods are included, while the objectives are different [17-18]. In detail, the objective of non-cooperative model is to maximize each stakeholder's profit such as Nash-type game and Stackelberg game, while the other one is to reduce the operation cost of the entire network such as the core and shapely value. Jadhav et al. [19] introduced a new pricing mechanism to solve the energy trading problem among the MGs using a noncooperative game theory-based algorithm, and both seller level game and buyer level game were considered. Zhang et al. [20] proposed a hierarchical

system architecture model based on Nash equilibrium to identify and categorize the key elements and technologies involved in P2P energy trading within MGs considering dwellings, offices and factories. To better study the trading behavior and provide guidance for a viable local market design, Chen et al. [21] presented different game theoretical frameworks including cooperative and non-cooperative games. Thomas et al. [22] proposed a game theoretic model for a local market design with storage sharing, using Nash equilibrium. However, these researches mainly involve the power research solely, the energy equipment considered is relatively simple, and do not involve more complex multi-energy coupling integrated energy system.

To solve this problem and deal with multi-type energy trading, Wang et al. [3] proposed a novel power-heating multilateral contract transaction mechanism considering spot market prices and established a Stackelberg game model for multi-master multi-slave based on the multi-lateral contract trading between integrated energy service providers and load converters. Wei et al. [23] proposed a new game model based on hierarchical Stackelberg game to solve the multi-energy transaction problem in the comprehensive energy system. However, this paper considered that each subject could only interact with the higher power grid, and did not involve the communication between subjects at the same level. Jing et al. [24] presented a concise solution to determine the fair prices for multi-energy trading based on a Nash-type non-cooperative game model between residential and commercial prosumers with guaranteed trading fairness. However, there is a lack of comparison study on running strategy and price determination mechanism under different trading modes [25].

DR programs can be incorporated in energy systems to reshape the load profiles and realize a load-shifting scheme for better energy performances [26-27]. Bozchalui et al. [28] established various equipment models in a residential building and found

that the DR may obviously enhanced the economic and environmental performance of energy systems. Wu et al. [29] proposed an improved optimization model for DR in a remote off-grid MG local on Dongfushan Island, China. Thomas et al. [30] presented an integrated tool to mitigate power quality issues in a MG considering different DR scenarios. These studies mainly focus on various kinds of electricity response [31]. However, despite electricity load shifting response, the substitution of different energy types in demand side can be also considered as a kind of ideal DR. In order to realize a comprehensive DR, Sheikhi and Bahrami [32] raised that the traditional DR mode cannot satisfy the more and more integrated energy systems, and firstly defined the integration of different energy types in demand side as integrated DR (IDR). Then, Good et al. [33] proposed an IDR model for multi-type loads. The results showed that the profit of multi-micro grid system and the comprehensive benefit of each user were improved by employing IDR. Zheng et al. [34] proposed an incentive-based IDR model for multiple energy carriers, the results demonstrated that the proposed model could achieve a win-win situation between multi-energy aggregator and consumers.

In short, according to the investigation of current studies, the following conclusions can be deduced:

- (1) Distributed energy trading problem has become a popular topic, but current studies mainly focus on one type of trading mode, a comparison study under different trading modes is necessary to deal with different scenarios.
- (2) Although many studies have considered multi-type energy trading problem, they are always limited to a single MEMG. And the multi-energy trading problem in a cluster need to be further explored.
- (3) Current studies mainly focus on end user's IDR solely, along with the occurrence of multi-level S/D relationship within a complicated energy structure,

multi-level multi-type IDR needs further study.

1.2. Contributions

To bridge the research gap, the main contributions of the paper are listed as follows:

(1) Multi-energy trading problems within a complicated MEMG cluster are studied and compared under two different scenarios: IA-based mode and direct trading mode without IA.

(2) Two methods based on SGT and S/D ratio are proposed to solve the price determination problem under the two scenarios.

(3) A multi-type multi-level IDR interacting with the timely changed energy trading price is implemented based on dynamic supply-demand two-way interaction. There are end users' IDR, MEMG supply response as well as energy interchange response among the MEMGs, respectively.

The remainder of this paper is organized as follows: Section 2 describes the problem definition; Section 3 establishes the bi-level models, especially, the energy trading price calculation methods under two situations; Section 4 extends the model to a scenario with three MEMGs in a cluster, and solves for the optimal operation modes, energy trading prices and IDR; and finally, the paper is concluded in Section 5.

2. Problem definition

2.1. Trading modes

When dealing with internal trading among the MEMGs in the cluster and external trading with the public energy network, a reasonable trading mode is necessary as shown in Fig. 1. As mentioned above, two kinds of trading modes are considered:

(1) IA-based trading mode

In this mode, IA acts as an independent entity to ensure private flow of information as well as fair dispatch of resources. Although the exchanged energy flows through

the established transmission infrastructure, all participants trade with IA directly with cash flows. IA is responsible for gathering necessary information, such as external energy price, surplus/insufficient energy amount of each MEMG, interest demand etc., and determining the internal energy trading prices accordingly.

(2) Direct trading mode without IA

This mode can refer Fig. 1 when deleting the red dotted line part. Each MEMG trades with neighborhoods directly without IA employment. Both of physical energy flow and cash flow are controlled by each MEMG operator.

It is necessary to indicate that the benefits of external energy network and demand side end users are not discussed in this study.

On the other hand, a multi-level multi-type IDR is proposed in this study other than traditional end users' electricity response solely. When a new trading price signal is delivered, MEMG cluster may optimize its end-users' energy using behavior, energy suppliers' operation strategy as well as internal energy share amount, which are named as end users' DR, MEMG supply response and MEMG cluster energy interchange response, respectively. In detail, end users' DR is to encourage customer to alter their load pattern to reduce or shift the peak demand according to the S/D relationship. MEMG supply response is to maximize its own benefit through the adjustment of energy generator operation, and MEMG cluster energy interchange response refers to encourage the MEMGs to adjust the energy sharing strategy with the neighborhoods.

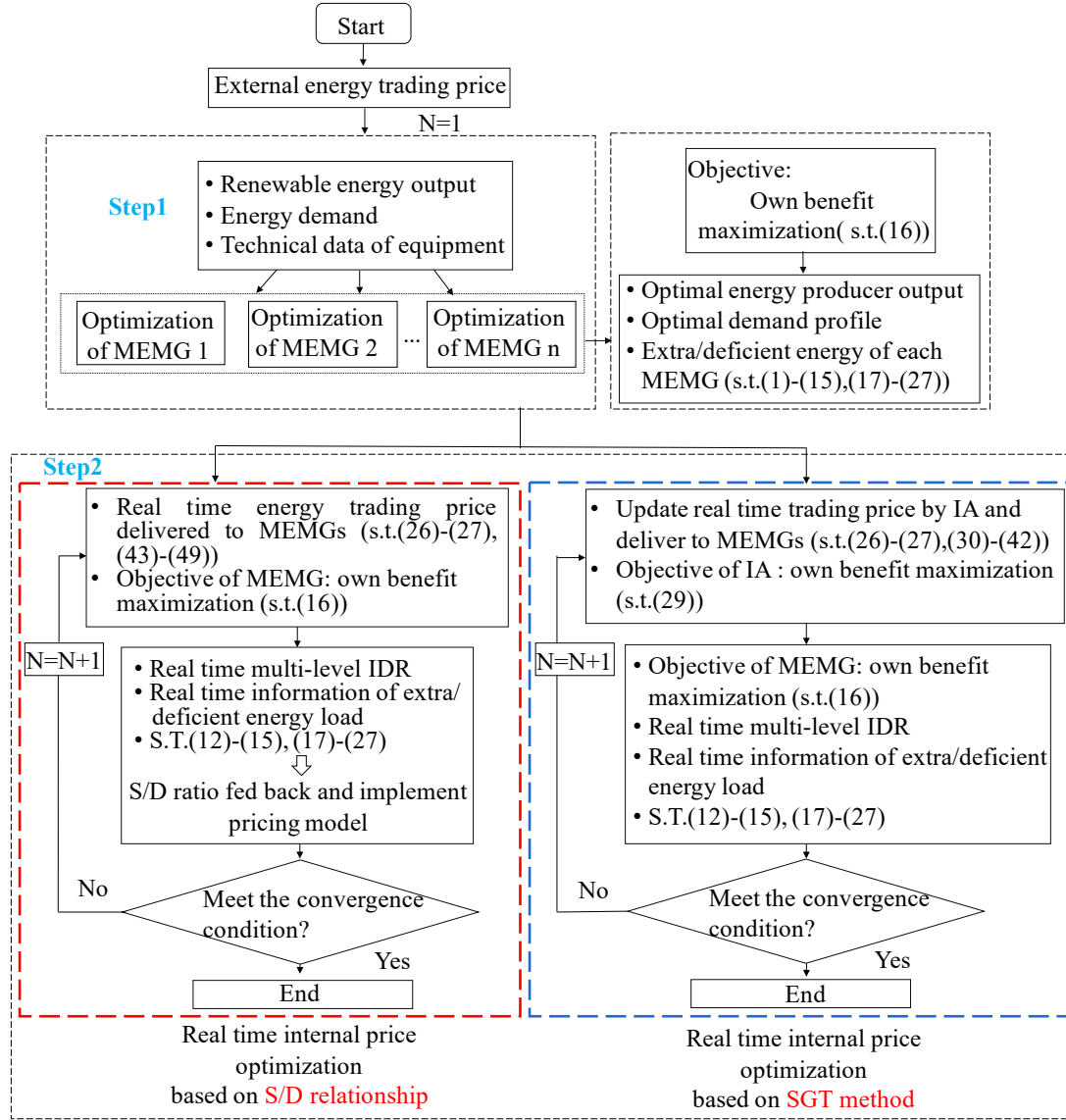


Fig. 2. Overall framework of this study.

3. Mathematical formulation

The mathematical formulations will be introduced in the following [35-36] which includes low-level model related to internal optimization of each MEMG, as well as up-level energy sharing model of nearby MEMGs which contains price determination methods based on SGT and S/D ratio. It is necessary to indicate that, to solve this problem, multi-time iteration between up-level and low-level model will be implemented since it is a nonconvex function.

3.1 Low-level model within each MEMG

3.1.1. Energy generator models

Each MEMG is combined of various energy generators in supply side and diversified load types in demand side which can be modeled as follows.

(1) CHP unit

Electricity generated by CHP unit can be calculated by multiplying consumed natural gas with calorific value and its generation efficiency as shown in Eq. (1). In addition, the energy generated is limited by its permitted maximum and minimum value. Here, L_{ng} is equal to 9.7 (kWh)/m³.

$$P_{mt,n,h} = M_{mt,n,h} L_{ng} \eta_{mt} \quad (1)$$

$$P_{mt,n,\min} \leq P_{mt,n,h} \leq P_{mt,n,\max} \quad (2)$$

In addition, heat recovered from CHP unit and finally useful heat is equal to electricity generated multiplied by heat/electric ratio, heat exchanger efficiency as well as backup boiler efficiency as follows:

$$P_{hc,n,h} = P_{mt,n,h} r_{mt} \eta_{wh} \eta_{hc} \quad (3)$$

(2) Gas boiler

Heat generated from gas-fired boiler can be calculated by multiplying gas consumed amount with calorific value heat and generation efficiency. In addition, the heat output must be within the capacity constraints, as shown in the Eq. (4) and Eq. (5).

$$Q_{gb,n,h} = M_{gb,n,h} L_{ng} \eta_{gb} \quad (4)$$

$$Q_{gb,n,\min} \leq Q_{gb,n,h} \leq Q_{gb,n,\max} \quad (5)$$

(3) Compression refrigerator

The cooling capacity of a compression refrigerator must be greater than the peak

demand of the cooling load divided by its performance coefficient and lower than the maximum allowable value, as shown in Eq. (6).

$$\frac{cl_{n,h}}{COP} \leq Cpc_{ec,n,h} \leq Cpc_{ec,n,\max} \quad (6)$$

(4) Battery constraints

The battery constraints can refer Eq. (7) and Eq. (8). The electricity charging and discharging of the battery cannot be implemented at the same time, and meanwhile cannot overcome the maximum values. $Bch_{n,h}$ and $Bdis_{n,h}$ are the charging and discharging power of the battery in nth MEMG at time h, respectively. Nch and $Ndis$ are the charging and discharging efficiency of the battery, respectively. $Es_{n,h}$ and $P_{bt,n,h}$ are the energy stored by the energy storage device and the net battery power of the nth MEMG at time h, respectively. $Es_n(0)$ and $Es_n(24)$ refer to the initial and final electricity reserved in the battery which are set as the same in this study and equal to 20% of the maximum value. $Bch_{n,\max}$ and $Bdis_{n,\max}$ are the upper limit of the maximum charge and discharge power, respectively; meanwhile, charging and discharging cannot be performed at the same time. $kech_{n,h}$ is 1 for charging state, and $kedis_{n,h}$ is 1 for discharging state.

$$Es_{n,h} = Es_{n,h-1} + Bch_{n,h}Nch - \frac{Bdis_{n,h}}{Ndis} \quad (7)$$

$$\left\{ \begin{array}{l} Es_n(0)=Es_n(24)=20\%Es_{n,\max} \\ Es_{n,\min} \leq Es_{n,h} \leq Es_{n,\max} \\ P_{bt,n,h} = Bch_{n,h} - Bdis_{n,h} \\ 0 \leq Bch_{n,h} \leq Bch_{n,\max} kech_{n,h} \\ 0 \leq Bdis_{n,h} \leq Bdis_{n,\max} kedis_{n,h} \\ kech_{n,h} + kedis_{n,h} \leq 1 \\ kech_{n,h}, kedis_{n,h} \in \{0,1\} \end{array} \right. \quad (8)$$

(5) PV unit

The installation area of the PV unit should not exceed the upper limit of the actual installation area of each producer and consumer, and the electricity produced should not exceed the rated generation capacity.

$$A_{PV,n} \leq A_{PV,n,\max} \quad (9)$$

$$0 \leq PV_{n,h} \leq A_{PV,n} I_{i_h} \eta_{PV} \quad (10)$$

(6) WT unit

Similar with PV unit, the actual power output of WT unit must also obey the allowable output range.

$$0 \leq P_{wt,n,h} \leq P_{wt,n,\max} \quad (11)$$

3.1.2. IDR of end users

In this study, despite electricity, thermal energy DR is also taken into consideration and named as integrated DR. As an aggregate of multi-energy flow coupling, MEMG has upper and lower limits on its cooling, heating and electric loads. Meanwhile, the total amount of cooling, heating and electric in one day before and after optimization must remain unchanged. The constraints are shown as follows:

$$\begin{cases} PEl_{n,h,\min} \leq ele_{n,h} \leq PEl_{n,h,\max} \\ PHet_{n,h,\min} \leq het_{n,h} \leq PHet_{n,h,\max} \\ PCol_{n,h,\min} \leq col_{n,h} \leq PCol_{n,h,\max} \end{cases} \quad (12)$$

$$\begin{cases} \sum_h PEl_{n,h} = \sum_h ele_{n,h} \\ \sum_h PHet_{n,h} = \sum_h het_{n,h} \\ \sum_h PCol_{n,h} = \sum_h col_{n,h} \end{cases} \quad (13)$$

Where $PEl_{n,h,\min}$, $PEl_{n,h,\max}$, $PHet_{n,h,\min}$, $PHet_{n,h,\max}$, $PCol_{n,h,\min}$ and $PCol_{n,h,\max}$ are the lower and upper limits of the electrical load, heat load and cooling

load of the n th MEMG at time h , respectively. $PEle_{n,h}$, $PHet_{n,h}$ and $PCol_{n,h}$ are the predicted electrical load, heat load and cooling load of the n th MEMG at time h , respectively. $ele_{n,h}$, $het_{n,h}$ and $col_{n,h}$ are the optimized electrical load, heat load and cooling load of the n th MEMG at time h , respectively. Eq. (13) ensures that the total electrical load, heat load and cooling load remain unchanged prior to and after load translation.

3.1.3. Energy balance constraints of each MEMG

The energy balance constraints of each MEMG are as follows which indicate that the energy output must be equal to the energy input in each time period. Here, cooling demand is transferred to electricity load since the adoption of compression chiller.

$$ele_{n,h} + \frac{col_{n,h}}{COP} + els_{n,h} + Bch_{n,h} = elb_{n,h} + Bdis_{n,h} + PV_{n,h} + P_{wt,n,h} + P_{mt,n,h} \quad (14)$$

$$hts_{n,h} + het_{n,h} = htb_{n,h} + P_{hc,n,h} + Q_{gb,n,h} \quad (15)$$

Where, $els_{n,h}$ and $elb_{n,h}$ are the electrical load sold/purchased by the n th MEMG at time h , respectively. COP is the refrigeration coefficient. $PV_{n,h}$, $P_{wt,n,h}$ and $P_{mt,n,h}$ are electricity generation from PV, wind turbine and micro-gas turbine of the n th MEMG at time h , respectively. $hts_{n,h}$ and $htb_{n,h}$ are the heat load sold/purchased by the n th MEMG at time h , respectively. $P_{hc,n,h}$ and $Q_{gb,n,h}$ are the output heating power of heat exchange device and gas boiler in the n th MEMG at time h , respectively.

3.1.4. Objective function of each MEMG

The objective function of each MEMG is to maximize the benefit in a certain time period as shown in Eq. (16). According to the current energy policy in China, generally, there are two operation modes for grid connected distributed energy system:

all energy generated on-site sold back to the utility grid and surplus part sold back to utility grid after self-satisfaction. Here, the latter one is considered.

$$\begin{aligned} \max BeMEMG_n = & \sum_h a_{n,h} \ln(1 + ele_{n,h} + \frac{col_{n,h}}{COP}) + \sum_h a_{n,h} \ln(1 + het_{n,h}) + \\ & \sum_h (Epes_h els_{n,h} - Epeb_h elb_{n,h}) + \sum_h (Hphs_{n,h} hts_{n,h} - Hphb_{n,h} htb_{n,h}) - \\ & \sum_h (Pg_{mt} M_{mt,n,h} + Pg_{gb} M_{gb,n,h} + C_{bt} (Bch_{n,h} + Bdis_{n,h})) \end{aligned} \quad (16)$$

$\sum_h a_{n,h} \ln(1 + ele_{n,h} + \frac{col_{n,h}}{COP}) + \sum_h a_{n,h} \ln(1 + het_{n,h})$ is the marginal benefits function

referring to energy charge paid by end users.

$\sum_h (Epes_h els_{n,h} - Epeb_h elb_{n,h}) + \sum_h (Hphs_{n,h} hts_{n,h} - Hphb_{n,h} htb_{n,h})$ is the energy

interchanged benefit paid by nearby MEMGs and external benefit trading with external energy network. Pg_{mt} and Pg_{gb} are the gas prices of micro-gas turbine and gas boiler, respectively. $M_{mt,n,h}$ and $M_{gb,n,h}$ represent the gas consumption of micro-gas turbine and gas boiler of the nth MEMG at time h, respectively. C_{bt} is the battery operating cost coefficient. Noting that, the operation cost must be removed from the benefit including primary energy cost, battery operating cost and etc.

3.2. Up-level model

The up level model contains the following parts: physical energy transmission constraints among the MEMGs, energy trading price constraints, as well as price determination methods.

3.2.1. Energy transmission constraints

In order to obey the basic trading principle, electricity and heat cannot be purchased and sold out simultaneously. In addition, the power exchanged among the entities should be lower than the maximum allowed amount. The relevant constraints are shown in Eq. (17)-(25).

$$nel_{n,h} = elb_{n,h} - els_{n,h} \quad (17)$$

$$0 \leq elb_{n,h} \leq elb_{n,\max} keb_{n,h} \quad (18)$$

$$0 \leq els_{n,h} \leq els_{n,\max} kes_{n,h} \quad (19)$$

$$keb_{n,h} + kes_{n,h} \leq 1 \quad (20)$$

$$nht_{n,h} = htb_{n,h} - hts_{n,h} \quad (21)$$

$$0 \leq htb_{n,h} \leq htb_{n,\max} khb_{n,h} \quad (22)$$

$$0 \leq hts_{n,h} \leq hts_{n,\max} khs_{n,h} \quad (23)$$

$$khb_{n,h} + khs_{n,h} \leq 1 \quad (24)$$

$$keb_{n,h}, kes_{n,h}, khb_{n,h}, khs_{n,h} \in \{0,1\} \quad (25)$$

Where $nel_{n,h}$ and $nht_{n,h}$ are internal electric and heat exchange power of the nth MEMG at time h, respectively. Meanwhile, $els_{n,h}$ and $elb_{n,h}$ cannot be performed at the same time. $elb_{n,\max}$ and $els_{n,\max}$ are the maximum values of the electrical power purchased/sold by the nth MEMG, respectively. $htb_{n,\max}$ and $hts_{n,\max}$ are the maximum values of the heat power purchased/sold by the nth MEMG, respectively. Again, $hts_{n,h}$ and $htb_{n,h}$ cannot be performed at the same time. Meanwhile, $keb_{n,h}$, $kes_{n,h}$, $khb_{n,h}$ and $khs_{n,h}$ are integer variables with a value of 0 or 1 for the nth MEMG at time h.

3.2.2. Basic principle of energy trading price determination

As mentioned above, this study will provide two kinds of energy trading price determination method under two situations: IA acts as an operator by employing SGT-based method, as well as direct trading based on S/D ratio method. In either case, to ensure the benefit of each stakeholder, the flowing constraints must be obeyed:

$$PrEs_h \leq Epes_h \leq Epeb_h \leq Preb_h \quad (26)$$

$$PrHs_h \leq Hphs_h \leq Hphb_h \leq PrHb_h \quad (27)$$

Where $Epeb_h$ and $Epes_h$ refer to the prices that MEMG purchases electricity from/sells back to IA/other MEMGs. Whereas, $Preb_h$ and $PrEs_h$ represent electricity prices trading with utility grid. Similarly, $Hphb_h$ and $Hphs_h$ represent the internal heating trading prices with IA/other MEMGs. $PrHb_h$ and $PrHs_h$ indicate the external heating trading prices with centralized energy network entity.

3.2.3. SGT method description in IA-based trading mode

When IA acts as an intermediate entity and aims at earn middle price difference, the competitive interest relationship between IA and MEMG cluster can be described as one leader and multi-follower in a SGT which can be described as follows:

$$F = \left\{ (IA \cup \{MEMG\}), \{Ame_{n,h} \cup Amh_{n,h}\}, \{Epes_h\}, \{Epeb_h\}, \{Hphs_h\}, \{Hphb_h\}, \{BeMEMG_{n,h}\}, \{BeIA_{n,h}\} \right\} \quad \forall n \in N \quad (28)$$

Eq. (28) shows the stakeholders, operation strategy and revenue function of game F . IA and $\{MEMG\}$ represent leader and followers in the game. $\{Ame_{n,h} \cup Amh_{n,h}\}$ represent surplus/insufficient amount of electricity and thermal energy of each MEMG respectively. $\{Epes_h\}, \{Epeb_h\}, \{Hphs_h\}, \{Hphb_h\}$ indicate the internal price strategy set made by IA. $\{BeMEMG_{n,h}\}, \{BeIA_{n,h}\}$ are the objective benefit functions of the trading entities. The equilibrium point of game F is $S = (Ame_{n,h}^* \cup Amh_{n,h}^*, Epes_h^*, Epeb_h^*, Hphs_h^*, Hphb_h^*)$, under which both IA and MEMGs cannot unilaterally change the price making and operation strategy for higher benefit.

Utility functions designed for IA and MEMG are in the following.

(1) Utility function of the leader

$$\begin{aligned} \max BeIA = & \sum_h (PrEs_h Exes_h - PrEb_h Exeb_h + PrHs_h Hxhs_h - PrHb_h Hxhb_h) + \\ & \sum_{n,h} (Epeb_h elb_{n,h} - Epes_h els_{n,h}) + \sum_{n,h} (Hphb_h htb_{n,h} - Hphs_h hts_{n,h}) \end{aligned} \quad (29)$$

$\sum_h (PrEs_h Exes_h - PrEb_h Exeb_h + PrHs_h Hxhs_h - PrHb_h Hxhb_h)$ refers to the middle

interest deduced by IA as an intermediate between centralized network and cluster.

$\sum_{n,h} (Epeb_h elb_{n,h} - Epes_h els_{n,h}) + \sum_{n,h} (Hphb_h htb_{n,h} - Hphs_h hts_{n,h})$ refers to the middle

interest deduced by IA as an intermediate between MEMG as a supplier and MEMG as a buyer.

Equations of energy transaction constraints are shown in the following:

$$\begin{cases} El_h = \sum_n els_{n,h} - \sum_n elb_{n,h} \\ Ht_h = \sum_n hts_{n,h} - \sum_n htb_{n,h} \end{cases} \quad (30)$$

$$\begin{cases} Exes_h = El_h, & El_h \geq 0 \\ Exeb_h = -El_h, & El_h < 0 \\ Hxhs_h = Ht_h, & Ht_h \geq 0 \\ Hxhb_h = -Ht_h, & Ht_h < 0 \end{cases} \quad (31)$$

In the formulas, El_h and Ht_h represent the amount of electricity and thermal energy transactions between IA and the external network at time h, respectively.

$Exes_h$ and $Exeb_h$ refer to the total amount of electricity sold by IA to the external network and the total amount of electricity purchased from the external network at time h. Similarly, $Hxhs_h$ and $Hxhb_h$ indicate the total amount of heat sold by IA to the external network and the total amount of heat purchased from the external network at time h, respectively.

(2) Utility function of the followers

Each follower targets at maximizing its own operating cost which has been shown

in Eq. (16). It should be noted that MEMG cannot buy and sell electricity/heat at the same time. As to the internal trading price determination, despite Eq. (26), other constraints must be obeyed, simultaneously. In the following, electricity is taken as an example for analysis. When one MEMG acts as an electricity purchaser, the deficient part it buys from IA can be described as:

$$elb_{n,h} = ele_{n,h} + \frac{col_{n,h}}{COP} + P_{bt,n,h} - PV_{n,h} - P_{wt,n,h} - P_{mt,n,h} \geq 0 \quad (32)$$

Under this situation, the load demand range of MEMG is as follows. Max indicates the permitted load demand after IDR.

$$PV_{n,h} + P_{wt,n,h} + P_{mt,n,h} - P_{bt,n,h} \leq ele_{n,h} + \frac{col_{n,h}}{COP} \leq \max \left(ele_{n,h} + \frac{col_{n,h}}{COP} \right) \quad (33)$$

The optimal electrical load demand can be obtained by calculating the first-order partial derivative of Eq. (16), that is $\frac{\partial(BeMEMG_n)}{\partial(ele_{n,h} + col_{n,h} / COP)} = 0$, and the optimal load is:

$$ele_{n,h} + \frac{col_{n,h}}{COP} = \frac{a_{n,h}}{Epeb_h} - 1 \quad (34)$$

The electricity price range can be derived by substituting Eq. (34) into Eq. (33):

$$\frac{a_{n,h}}{\max(ele_{n,h} + col_{n,h} / COP) + 1} \leq Epeb_h \leq \frac{a_{n,h}}{PV_{n,h} + P_{wt,n,h} + P_{mt,n,h} - P_{bt,n,h} + 1} \quad (35)$$

It can be known from Eq. (35) that, for any electricity purchaser MEMG in the group, the trading price corresponds to a specific range and is abbreviated as follows:

$$Epeb_h \in [Epeb_{n,h,\min}, Epeb_{n,h,\max}] \quad (36)$$

Noting that if the electricity price is in the range of $Epeb_h \leq Epeb_{n,h,\min}$, the optimal load demand of electricity is $\max(ele_{n,h} + col_{n,h} / COP)$. Otherwise, if $Epeb_h \geq Epeb_{n,h,\max}$, the optimal load demand is $(PV_{n,h} + P_{wt,n,h} + P_{mt,n,h} - P_{bt,n,h})$. That is,

the load is related to the price, and has upper and lower limits.

Similarly, when MEMG acts as a seller, the optimized range of its electric load demand and the corresponding range of electricity price can be deduced as follows:

$$\min(ele_{n,h} + \frac{col_{n,h}}{COP}) \leq ele_{n,h} + \frac{col_{n,h}}{COP} \leq PV_{n,h} + P_{wt,n,h} + P_{mt,n,h} - P_{bt,n,h} \quad (37)$$

$$\frac{a_{n,h}}{PV_{n,h} + P_{wt,n,h} + P_{mt,n,h} - P_{bt,n,h} + 1} \leq Epes_h \leq \frac{a_{n,h}}{\min(ele_{n,h} + col_{n,h} / COP) + 1} \quad (38)$$

Eq. (38) can be simplified as $Epes_h \in [Epes_{n,h,\min}, Epes_{n,h,\max}]$.

Similarly, in the case of thermal energy, the optimal thermal load demand can be obtained through partial derivative of heat load. The corresponding heat load demand range, heat purchasing and sale price range can be obtained, as shown in the following:

$$P_{hc,n,h} + Q_{gb,n,h} \leq het_{n,h} \leq \max(het_{n,h}) \quad (39)$$

$$\frac{a_{n,h}}{\max(het_{n,h}) + 1} \leq Hphb_h \leq \frac{a_{n,h}}{P_{hc,n,h} + Q_{gb,n,h} + 1} \quad (40)$$

Eq. (40) can be abbreviated as $Hphb_h \in [Hphb_{n,h,\min}, Hphb_{n,h,\max}]$.

$$\min(het_{n,h}) \leq het_{n,h} \leq P_{hc,n,h} + Q_{gb,n,h} \quad (41)$$

$$\frac{a_{n,h}}{P_{hc,n,h} + Q_{gb,n,h} + 1} \leq Hphs_h \leq \frac{a_{n,h}}{\min(het_{n,h}) + 1} \quad (42)$$

Eq. (42) can be abbreviated as $Hphs_h \in [Hphs_{n,h,\min}, Hphs_{n,h,\max}]$.

3.2.4. S/D ratio method description in direct trading mode

As to the price mechanism depending on S/D ratio of the whole MEMG cluster under direct trading mode without IA, the objective function is the same as that of SGT: each MEMG targets at maximizing its own operating benefit referring Eq. (16). According to the S/D relationship, there exists the following situations:

(1) When $\sum_n elde_{n,h} = \sum_n elsurp_{n,h} = 0$, no internal trading among the MEMGs.

Where $\sum_n elde_{n,h}$ and $\sum_n elsurp_{n,h}$ are the total power demand and total power supply of the MEMG cluster at time h , respectively.

(2) Either $\sum_n elde_{n,h}$ or $\sum_n elsurp_{n,h}$ is equal to 0, the deficiency/surplus part can be traded with external energy network.

$$\left\{ \begin{array}{l} \sum_n elde_{n,h} = 0, \sum_n elsurp_{n,h} \neq 0, Pes_h = PUs_h \\ \sum_n elsurp_{n,h} = 0, \sum_n elde_{n,h} \neq 0, Peb_h = PUb_h \end{array} \right\} \quad (43)$$

Where Pes_h and Peb_h are the internal electricity sale/buy-back price without IA at time h , respectively. PU_s_h and PU_b_h are the utility electric sale//buy-back price at time h , respectively.

(3) when $\sum_n elde_{n,h} = \sum_n elsurp_{n,h} \neq 0$, for fairness,

$$Pes_h = Peb_h = \frac{PU_s_h + PUb_h}{2} \quad (44)$$

(4) when $\sum_n elde_{n,h} \geq \sum_n elsurp_{n,h}$, assuming $k_h = \frac{\sum_n elsurp_{n,h}}{\sum_n elde_{n,h}}$, $0 \leq k_h < 1$ in this

situation. As is known, the internal trade price is inversely proportional to S/D ratio which can be expressed in Eq. (45).

$$Pes_h = \frac{1}{ck_h + d}, \quad c > 0 \quad (45)$$

Where k_h is supply/demand ratio at time h . c and d are the solution parameters. Since the price is a continuous function, thus:

$$Pes_h = \begin{cases} 1/d = PUs_h, & k_h = 0 \\ 1/(c+d) = (PU_s_h + PUb_h)/2, & k_h = 1 \end{cases} \quad (46)$$

By calculating Eq. (46), c and d can be deduced, and then Pes_h can be calculated as:

$$Pes_h = \frac{PUes_h(PUes_h + PUEb_h)}{PUes_h(1 + k_h) + PUEb_h(1 - k_h)} \quad (47)$$

Economic balance must be obeyed which can be expressed as follows:

$$\sum_n elsurp_{n,h} \cdot Peb_h = \sum_n elde_{n,h} \cdot Pes_h - (\sum_n elde_{n,h} - \sum_n elsurp_{n,h}) \cdot PUes_h \quad (48)$$

Thus, Peb_h can be deduced as:

$$Peb_h = \frac{1}{k_h} \cdot PIAes_h - (\frac{1}{k_h} - 1) \cdot PUes_h \quad (49)$$

Similarly, the internal energy trading price can be deduced using the same approach when $k_h \geq 1$.

4. Numerical study

In order to certify the feasibility of the proposed model in this study, a micro cluster combined of three park-level MEMGs is studied [37-39]. The model is solved by using YALMIP/CPLEX optimization solver based on MATLAB platform [40-41].

4.1. Description of the case study

Here, three park-level MEMGs namely MEMG1, MEMG2, and MEMG3 will be considered in this study. It is assumed that both the dispatchable and non-dispatchable distributed generators exist in the networked MEMGs. Each MEMG contains a battery energy storage system (BESS), controllable distributed generator, renewable distributed generator, and end users' load. If the energy inter produced cannot realize self-sufficient of a MEMG, nearby MEMGs or the centralized energy system will supply the deficient part, and vice versa.

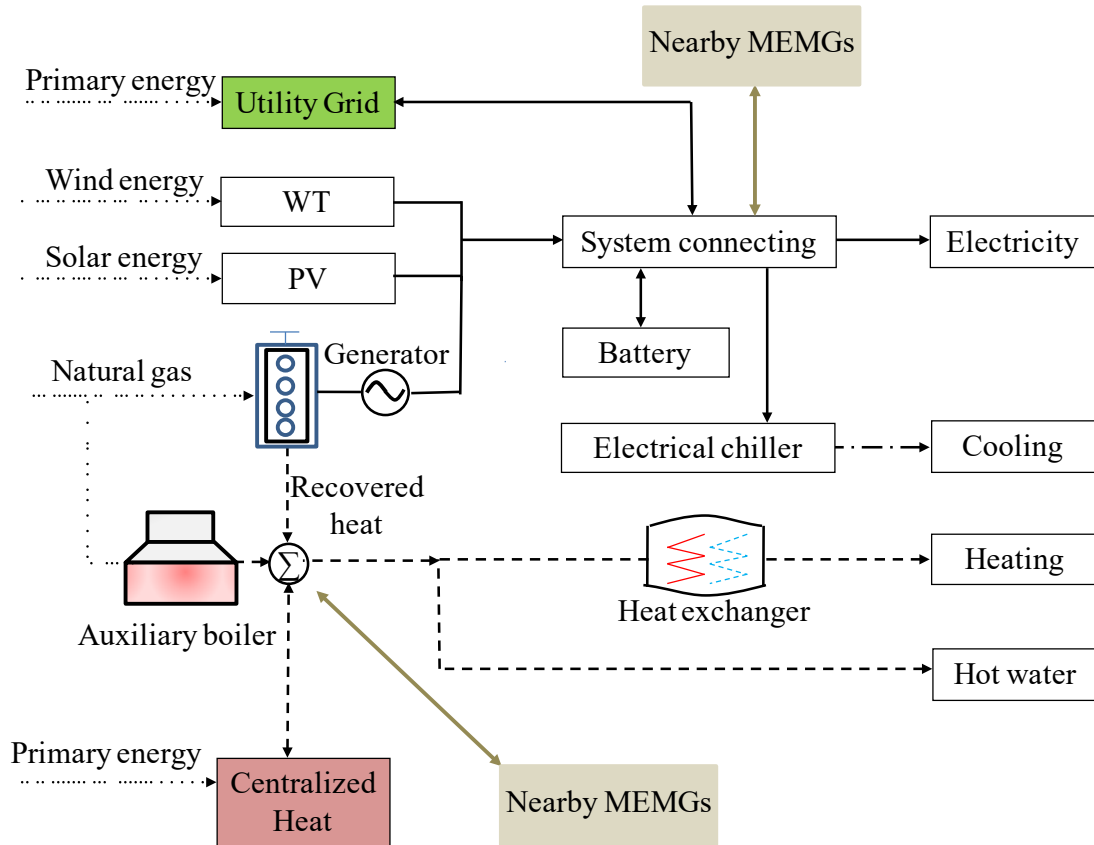


Fig. 3. Energy flow structure of each MEMG.

4.2. Necessary assumption and explanation

As one of the most important inputs, various technical assumptions are listed in Table 1 including technology efficiency of employed energy generators. It should be noted that the adjustment range of the controllable multiple loads is set as $\pm 20\%$ of the load forecast curve, and the operation cost coefficient of energy storage is 0.02 CNY/kWh. On the other hand, Table 2 shows the generators' capacities employed in each MEMG. Generally, MEMG2 enjoys the highest capacities for PV, WT and CHP units, while the battery capacities are set as the same in all three parks.

Table 1. Technical assumptions [42-44].

Items	Technology Efficiency
Micro gas turbine	30 %
Heat to electricity ratio	1.47

Natural gas boiler	90 %
Heat recovery boiler	80 %
Heat exchanger	90 %
Battery charging	95 %
Battery discharging	95 %
Limits of battery storage energy	10 % - 100 %
Compression chiller	4.7
PV	14%

Table 2 Capacity of each generator.

MEMG	PV/kW	WT/kW	CHP/kW	Gas boiler/kW	Battery/kWh	Electrical chillers/kW
1	100	100	100	150	100	100
2	650	550	200	120	100	80
3	100	100	100	120	100	120

Generally, the energy price categories can be divided into two parts: internal energy trading price depending on the optimization result, as well as external price trading with public energy network which is shown in Table 3. Due to the adoption of time of use price mechanism, the electricity price made by utility grid is different in various time periods. In addition, in order to promote the adoption of CHP unit, the natural gas price may enjoy preferential treatment while the auxiliary boiler cannot enjoy the same price discount. Moreover, as to the buy-back energy price, the determination is always different from case to case, and is highly dependent on negotiation among related stakeholders. Here, buy-back price is determined referring [45-46].

Table 3. Energy prices for analysis.

Energy type	Time period division	Unit price
Electricity price made by utility entity (CNY/kWh)	8:00-11:00	1.189
	13:00-16:00	
	18:00- 22:00	
	7:00-8:00	0.738
	11:00-13:00 16:00-18:00	
Electricity buy-back price (CNY/kWh)	00:00-7:00	0.423
	22:00-24:00	
Heat price made by centralized system (CNY/kWh)	0:00-24:00	0.352
Heat buy-back price (CNY/kWh)	0:00-24:00	0.620
Gas price for CHP (CNY/m ³)	0:00-24:00	0.352
Gas price for gas boiler (CNY/m ³)	0:00-24:00	2.700
	0:00-24:00	3.870

Fig. 4 shows the solar irradiation and wind speed data used in this study [47]. The weather data in this paper comes from a typical summer day in Shanghai in July. Generally, the two sets of data may complement each other in a whole day, since solar irradiation peaks at PM 1:00 in daytime which is 0.55 kW/m², whereas wind speed peaks at PM 11:00 and 12:00 in nighttime which is as high as 7.31 m/s.

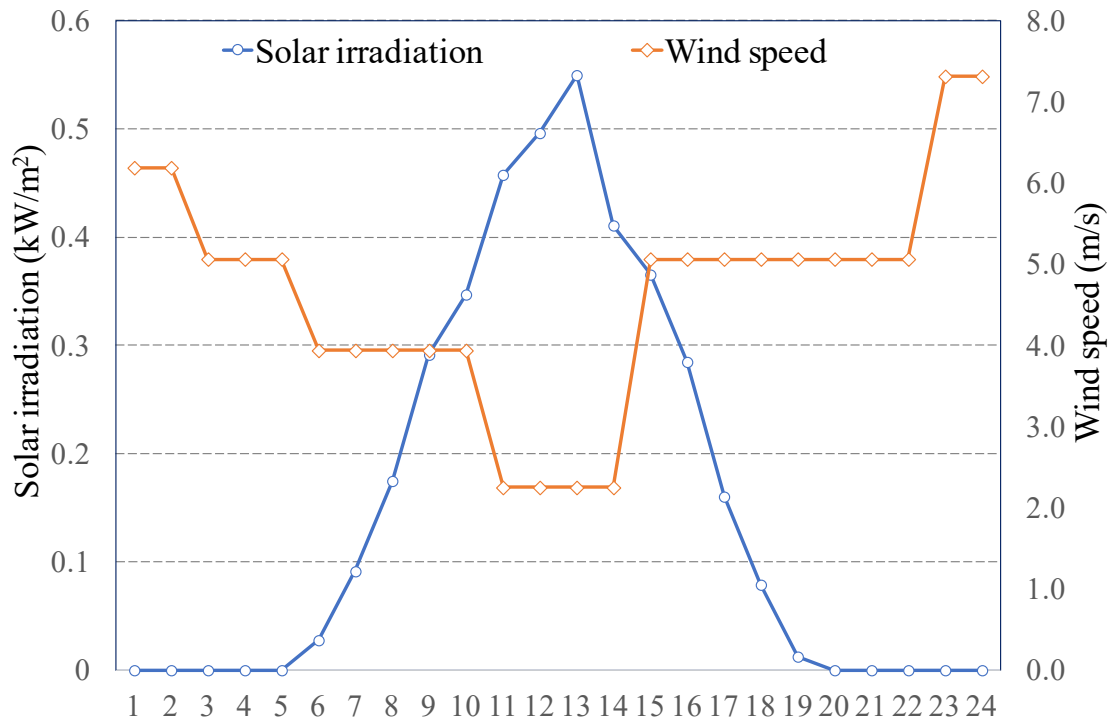


Fig. 4. Solar irradiation and wind speed inputs.

4.3. Results and discussions

By employing the proposed model, despite the internal energy trading price determination, hourly energy operation strategy, economic performance, as well as IDR under each scenario can be deduced. Here, for simplicity, a typical day in summer has been selected for detail analysis.

4.3.1. Optimal operation strategy of each MEMG

As an example, Fig. 5 shows the electricity balance of MEMG1, MEMG2 and MEMG3, respectively when energy is interchanged with each other in the neighborhoods using SGT method employing IA. It can be found that, generally, MEMG2 acts as an electric seller within the combined cluster, whereas both MEMG1 and MEMG3 act as electric buyers. This is mainly due to the relatively high capacity of district energy generators adopted in MEMG2 referring table 2, and low ones adopted in MEMG1 and MEMG3. That is to say, only if there exist relatively large differences between energy supply and demand sides in each MEMG, the energy

interchange among the neighborhoods can be realized. In addition, although CHP unit is applied in MEMG2, it only works for few hours due to the relatively high operation cost compared with PV and wind system from economic aspect. Moreover, it can be found that CHP units of MG1 and MG3 always operate during daily time, this is because during this time period, both electricity and heat demands are relatively high, the economic performance of CHP is better since it may generate electric and thermal energy, simultaneously. On the contrast, thanks to the relatively low electric price during nighttime, the MEMGs prefer to buy electricity directly from external public energy network other than produce it themselves. Again, the load difference between nighttime and daytime of MEMG3 is the largest, CHP unit and internal interchanged energy may satisfy its daily electric demand due to its relatively low renewable energy installation, whereas wind power satisfies its energy demand during nighttime.

On the other hand, as to the heat balance of the three MEMGs after optimization. Compared with the situation of electricity, the running strategy is relatively simple. Heat demand is mainly supplied by CHP unit, gas boiler and heat exchange or a combination of them. Here, MEMG1 acts as a heat buyer, whereas both MEMG2 and MEMG3 act as heat sellers. As to MEMG1, different from the situation of electricity, the interchanged part occupies few of total heat demand. This is because, during summer period, total heat demand is relatively low compared with electric demand which also contains the electric consumption that for cooling loads, and thus most of the heat demand of MEMG1 can be supplied by CHP unit and gas boiler of its own. In addition, as to MEMG2 and MEMG3, the surplus part of heat generated from CHP unit after self-satisfactory will be sold out to IA for additional benefit. It is worth noting that, the sum of surplus part supplied by MEMG2 and MEMG3 is higher than the insufficient part of MEMG1, which means part of surplus heat is sold out to the

external heat network.

The optimization result of electric and heat balance employing S/D ratio method can be also deduced under direct trading mode, which will not be described in detail.

4.3.2. IDR analysis

Fig. 6 shows the load shift and transfer status before and after the optimization in both trading modes. It can be found that, the loads change a lot in all three MEMGs compared with the initial value. Generally, since the load response and energy price are optimized simultaneously, each MEMG tends to lower the load demand when the energy price is high and vice versa. That is to say, each MEMG wants to maximum its own benefit in each time period through the dynamic price and load interaction.

In addition, it can be found that, although the load change trend is similar in two situations, while due to the different internal exchange price determination principle, the load fluctuation utilizing S/D ratio-based method under direct trading mode is more dramatic and totally different from the situation that uses SGT method employing IA. This is because the internal energy exchange price is totally depending on the S/D ratio, the load DR is more sensitive.

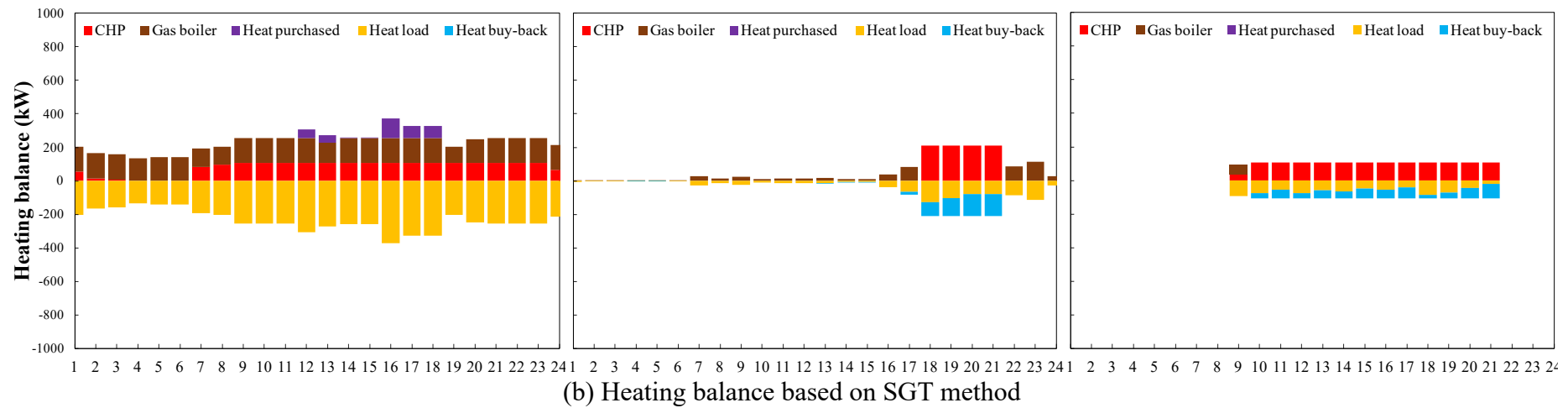
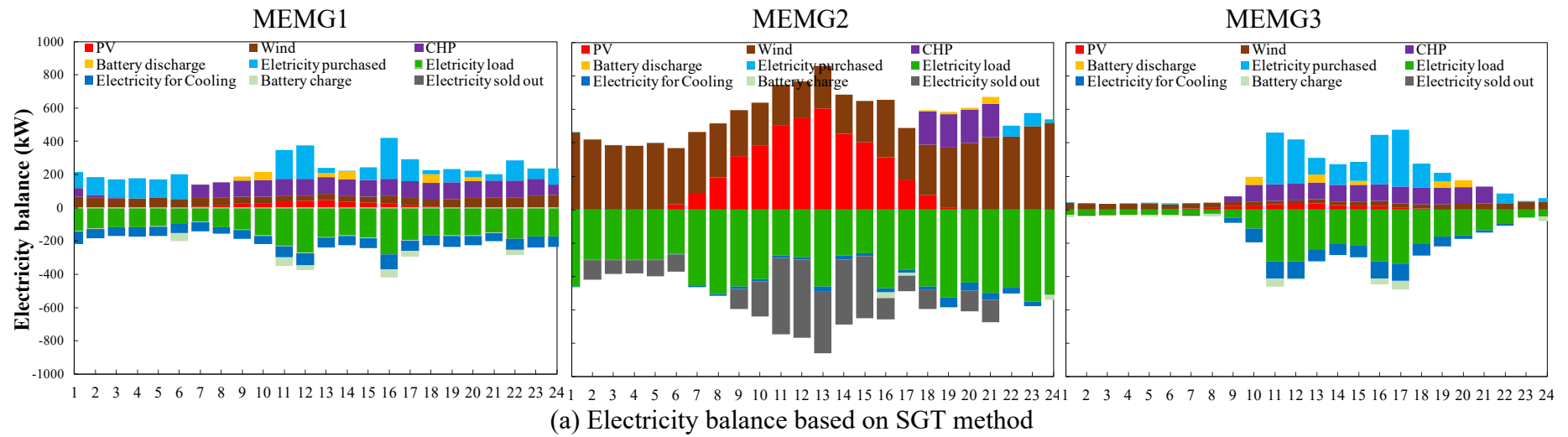


Fig. 5. Electricity and heating balances of the MEMG cluster employing IA.

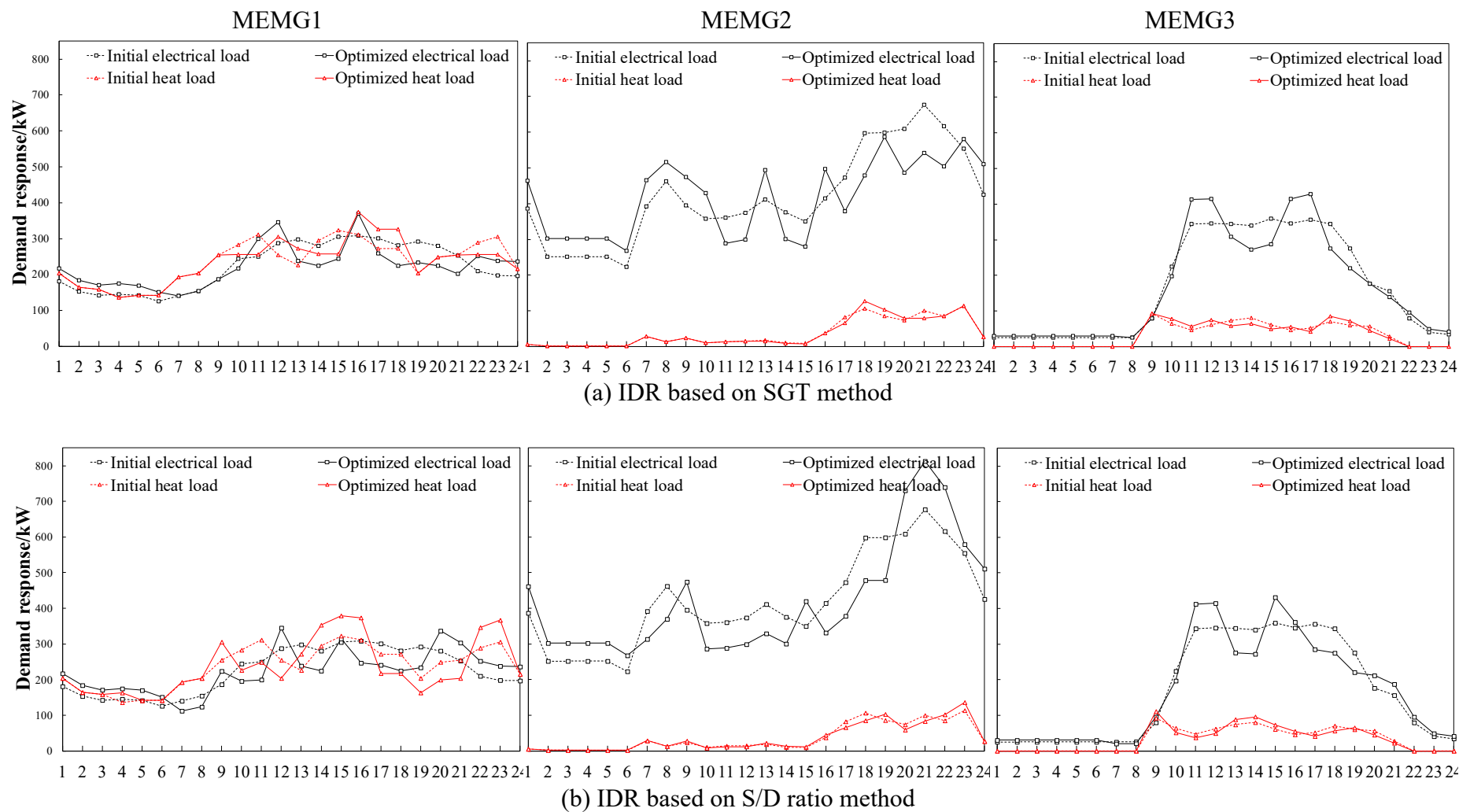


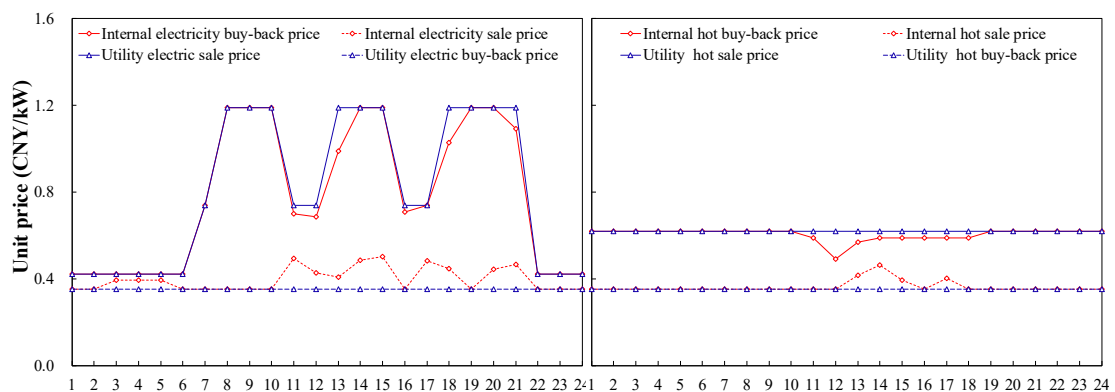
Fig. 6. IDR of each MEMG.

4.3.3. Simulation results of energy trading price

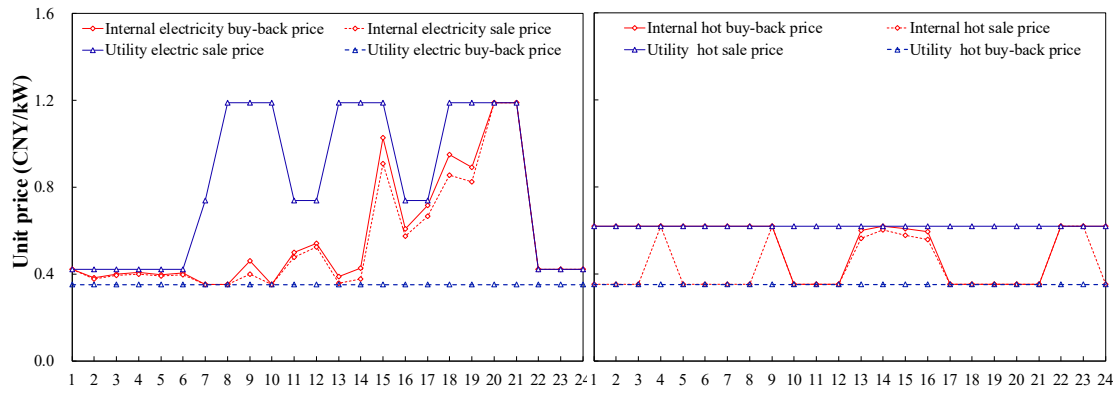
The deduced optimal energy trading prices by SGT method employing IA are shown in Fig. 7 (a). It can be found that, generally, the internal energy trading prices are between the value of sale and buy-back prices made by external utility grid in all time periods. To be specific, the utility energy sale price is highest, followed by internal sale price, internal buy-back price, and utility buy-back price, respectively. As to electricity, the trading prices made by IA and utility grid are almost the same during 0:00-6:00 and 22:00-24:00 obviously. This is because, during this time period, the electricity sale and buy-back price made by utility grid are relatively low and the profit margin of IA is limited. Whereas, during 10:00-22:00, the differences are relatively large. As to thermal energy, generally, similar with electricity, the external heating sale price is highest, followed by internal heating sale price, internal heating buy-back price and external heating buy-back price, respectively. In addition, it can be concluded that during 0:00-10:00 and 19:00-24:00, since there is no heating exchange among the MEMGs, the internal heating price is optimized as the same as external one. While, during 11:00-18:00, the internal heating buy-back price is higher than the external one, and the internal heating sale price is lower than the external one to promote MEMGs exchange surplus/insufficient energy with neighborhoods firstly other than the external supplier, which can increase the stability and flexibility of the entire energy system. Moreover, it is necessary to indicate that the prices made by IA under this scenario change slightly in order to encourage the MEMGs exchange energy internally other than with external energy network directly.

On the other hand, when each MEMG trades directly, as aforementioned, S/D ratio-based method can solve the problem. The optimized internal energy trading price can refer Fig. 7 (b). Similarly, the internal price is also in the range of external ones,

which is the same as the simulation result when using SGT method. However, the difference between internal buy-back and sale price is relatively low and the determined prices change more extremely compared with SGT-based method. This is because under this scenario, the internal energy trading price is totally depending on the timely changed S/D ratio. For example, during 8:00-14:00, since the S/D ratio is more than 1, the internal energy trading price is relatively low, the MEMGs belongs to electricity-sale cluster, the surplus electricity will be sold out to the utility grid for more benefit. Whereas, the situation is opposite during 15:00-20:00, at this time period, the S/D ratio is less than 1, and the internal price is relatively high due to the internal energy deficiency and the deficient part must be supplied by the external grid. Moreover, during 22:00-24:00, since there is no energy trading among the MEMGs, all deficient part is satisfied by external grid, the energy price is the same as external ones. In a word, in the S/D ratio-based method, the internal price reflects the relationship between supply and demand directly, high internal energy trading price indicates that during this time period, the whole cluster is in a state of energy deficiency, which will encourage the MEMGs lower load self-using and produce more energy for interchange.



(a) SGT-based method considering IA's benefit



(b) S/D ratio-based method under direct trading without IA

Fig. 7. Optimal price determination of internal energy exchange.

4.3.4. Total energy trade amount of the whole cluster

Fig. 8 shows the energy trade amount between external centralized network and the whole MEMG cluster in three scenarios. Generally, it can be found that, compared with (Independent operation) IOP in which each MEMG optimized individually, the combination of cluster can reduce the total energy trade amount with external centralized network obviously both for energy purchased and energy buy-back part, especially under the SGT-based scenario employing IA. To be specific, total energy exchange amount reduce 58% and 37% under IA employment and direct trading scenarios, respectively. That is to say, from the entire energy aspect, through the positive coalition, the proposed three MEMGs exchanged a large amount of energy with each other, and balanced surplus and deficient part of energy internally. However, it is worth noting that, when focusing on electricity and heat respectively, the situation is different. As to electricity, the conclusion is the same as total energy trade trend in all scenarios, whereas as to heat, the trading amount with centralized heat network of MEMG cluster under direct trade is larger than IOP scenario both for purchased and buy-back part. This is because, without IA, the MEMGs will try their best to maximize profits themselves and have no concerns on the interest loss and energy source. On the contrast, when employing IA, since the benefit will be shared with IA

inevitably and the price difference is larger when trading with external network, the MEMGs will balance energy S/D through internal exchange as far as possible due to the relatively low internal price gap.

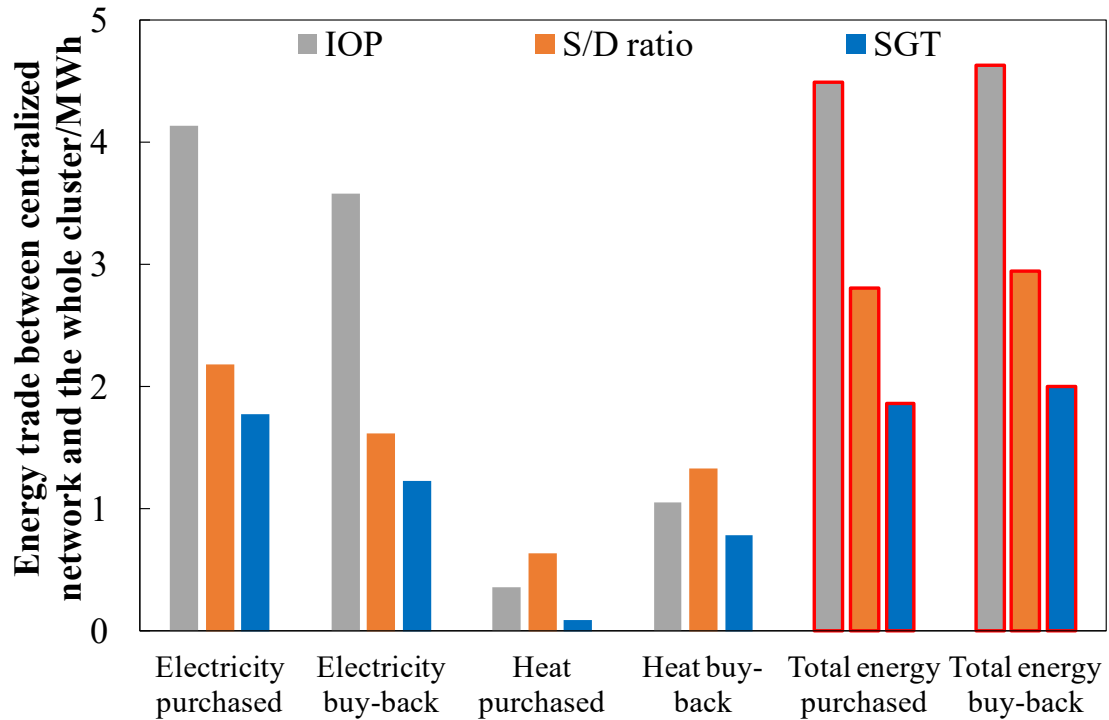


Fig. 8. Energy trade amount between centralized network and the whole cluster.

4.3.5. Benefit analysis of each participant in different scenarios

Fig. 9 shows the hourly benefit optimized of each MEMG in three scenarios in a typical day. Generally, all MEMGs may earn more benefit during daytime than nighttime for either one of the studied operation modes. In addition, it can be concluded that although in majority of the time, the combination of MEMG may deduce the highest benefit of the whole cluster without considering IA, the result is not constant in all hours. For example, during the time period of PM 8:00 – PM 10:00, MEMG obviously gains more benefit under the scenario employing IA than other two modes. This is because the optimization object is to maximize the whole day's benefit other than each hour.

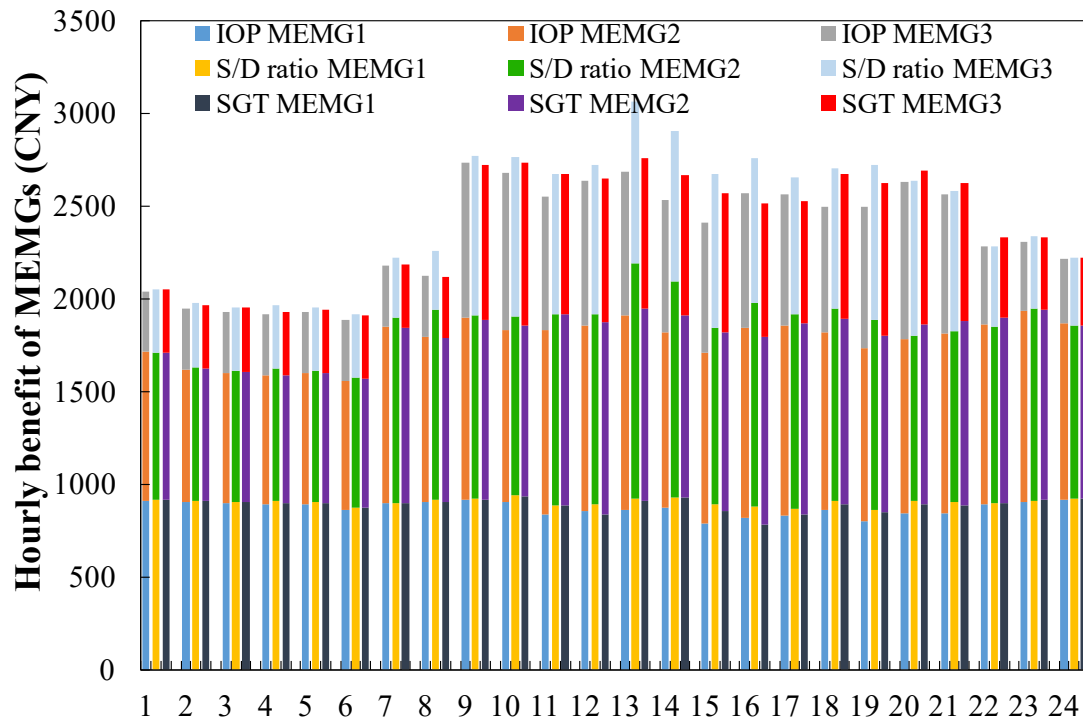


Fig. 9. Hourly benefit of each MEMG in three scenarios in a typical day.

Fig. 10 shows the hourly benefit deduced by IA in a typical day employing SGT-based method. It can be concluded that, through the utilization of SGT method, the profits of the leader-IA and the multiple flower-MEMGs can be balanced. Obviously, only if either energy interchange among the MEMGs or external grid exists, the benefit of IA is more than 0, the more the higher. In addition, since the peak time of energy demand is during daytime, the energy exchange amount is largest during this time period and IA can get higher profit especially from AM 11:00 to PM 6:00. Whereas, during AM 1:00, AM 7:00 - AM 10:00, PM 7:00, PM 10:00 - PM 12:00 periods, the benefit of IA is equal to 0 since each MEMG can realize energy self-sufficiency without help from outside.

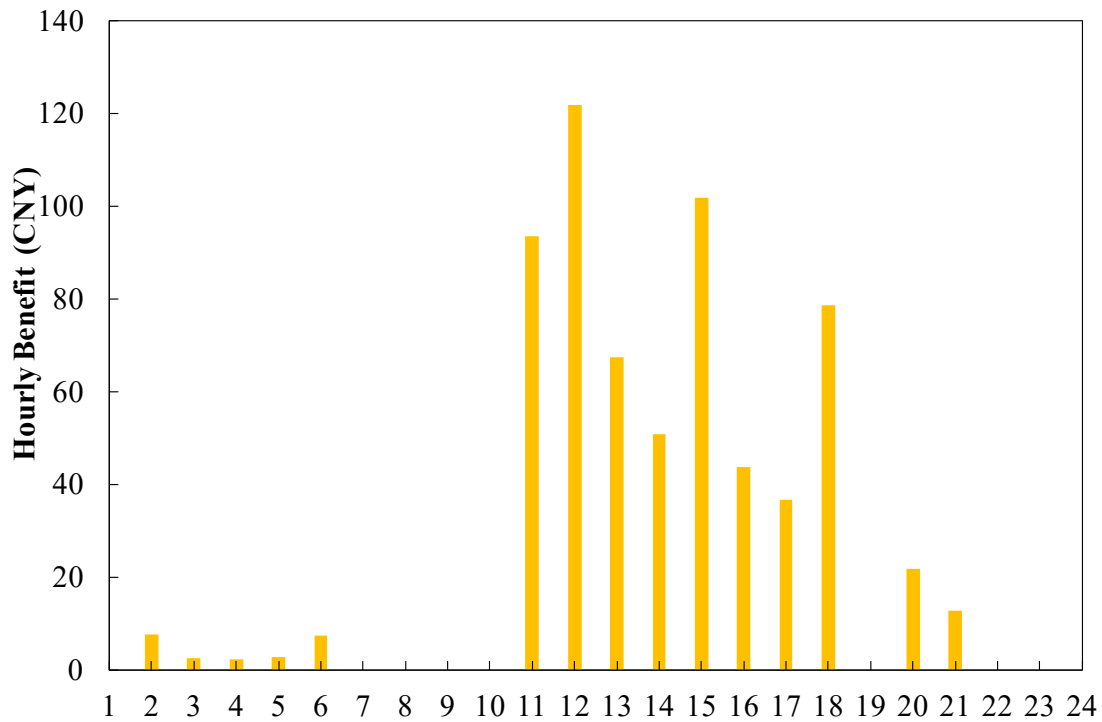


Fig. 10. Hourly benefit of IA in a whole day utilizing SGT method.

Table 4 shows the benefit of each stakeholder in a typical day. It can be found that, generally, MEMG2 has the highest profit, followed by MEMG 1 and MEMG 3, respectively. Again, compared with the situation of IOP where all MEMGs optimized independently, benefit of each MEMG increases through energy interchange among them no matter IA is employed or not. In addition, under direct trading mode without IA, each MEMG may derive more benefit since the avoidance of interest sharing with IA. It also can be seen that, obviously, even the benefit of IA is taken into consideration, it is far less than each MEMG. This is because the profit deriving modes of the two kinds of stakeholders are totally different. The IA's profit relies on the reselling of heat and electric energy exchanged among the MEMGs/external grid rated to exchanged amount and price gap. Whereas, each MEMG's profit depends on the energy charge paid by end-users. To be specific, when considering/ignoring IA, the profits of MEMG1, MEMG2, MEMG3 and whole MEMG cluster increase 1.77%, 1.17%, 2.71%, 2.92% as well as 3.68%, 4.30%, 5.51%, 4.36%, respectively.

Table 4 Benefit analysis in a typical day.

Stakeholders	Benefit (CNY)		
	IOP	Without IA	With IA
MEMG1	20914	21683	21285
MEMG2	21753	22688	22007
MEMG3	13627	14378	13997
MEMG cluster	56294	58749	57289
IA	/	/	651

5. Conclusions and policy implications

This study provides a bi-level model containing different price determination approaches for a MEMG cluster under different trading modes. According to the simulation results, the following conclusions can be revealed:

(1) The proposed MEMG group effectively improves the flexibility of the regional energy supply system, and can realize better overall economic performance through complementary and interworking. Compared with the situation that each MEMG acts alone, the combination of a cluster obviously increases the overall benefit of the whole cluster, which are 1.8% and 4.4% for IA trading mode and direct trading mode in the studied typical day, respectively. The result encourages policy makers to develop more tailored policy instruments to encourage such kind of internal trading.

(2) The price deduced employing S/D ratio-based method changes more intense than the situation that uses SGT method, and its internal energy trading price gap between energy sale and buy-back is relatively small. This is because, when employing IA, more price gap is preferred for more benefit. Meanwhile, the dynamic price change causes positive load shifting both for electricity and heating demand for

overall benefit maximization.

(3) The proposed SGT-based approach and S/D ratio-based approach are effective to deal with the internal energy trading price determination problem under two scenarios. Correspondingly, total energy trading amounts with external grid reduce 58% and 37%, respectively. That is to say, although the total benefit of MEMG cluster is higher under direct trading mode, the internal energy self-balance level is lower. In a word, when seeking the self-sufficient ability and stable energy demand, it is suggested to employ IA, whereas if the maximum benefit is paid more attention, direct trading without IA is preferred.

This study may help policy makers to more accurately predict merits and demerits associated with different energy trading modes among MEMGs. In the following studies, not only the internal energy trading price, but also the external price made by external network will be considered as market-oriented variables timely changed affecting S/D relationship and energy trading strategy in the market. In addition, the profit allocation within a MEMG to the prosumers/consumers needs further consideration.

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